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A NEW PROPOSAL ARCHITECTURE FOR REFERENCE PLE: REFPLE

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ABSTRACT

The purpose of this paper is to design reference architecture for intelligent personal learning environments (PLE). This research effort is a result of a multi-stage approach. First, semi-structured interviews have been conducted with software development practitioners working in a large technology and communications services provider company. Interview content has been evaluated using thematic analysis. Second, reference architecture for intelligent personal learning environments has been designed and validated using the Software Architecture Analysis Method (SAAM). During SAAM evaluation, the output of the first phase and previous Personal Learning Environment literature reviews has been used. As a result, we demonstrated a novel architecture for personal learning environments. Proposed personal learning architecture has met the requirements of industry demands and previous literature.

Keyword head: Personal Learning Environment, Software Architecture, ICT Professionals, Interview

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1. INTRODUCTION

Today, professionals working in large scale enterprise organizations, entrepreneurs in start-ups and companies do not only compete with other individuals and companies. Industrial Internet of Things and Industry 4.0 powered by Artificial Intelligence (AI) are becoming important factors in disrupting their traditional business models and professions. According to a research published by World Economic Forum, physical or cognitive jobs that can be automated will eventually be replaced by software and robots and growing use of machine intelligence in an industrial internet

of things world will impact the skills set of future human workforce in a radical manner [1]. In a recent study, Machine Learning researchers have argued that AI performance will exceed humans' for all tasks in 45 years and all human jobs will be fully automated in 120 years, with Asian participants more optimistic than North Americans about the dates [2]. Even in the present time, this paradigm shift in industry is visible as an important skills shortage. In a recent OECD report, at least 40 % of workers in OECD countries stated that their competence is not sufficient for the jobs they are interested in and companies consistently experience difficulties in filling needed jobs [3]. In another industrial survey conducted by Manpower Group, 18.000 employers in 43 countries were asked about the impact of automation on headcount in the near future and they've found that more than 90 % of the employers expect an impact on their enterprise due to digital transformation [4]. Therefore immediate action should be taken to up skill and reskill employees [4]. Employees' skills need to be developed to take advantage of Artificial Intelligence [5]. An interesting stackoverflow.com study shows that software developers in developed countries use Python and R programming languages much more often, which means they've already started investing more in Artificial Intelligence and Machine Learning related technologies [6]. In such an environment, traditional methods and platforms for training organizations and individuals are not sufficient any more. The aforementioned learning tradition that assumes learning takes places in a school classroom is not new and goes at least 2500 years back, to the Confucius era [7]. Toffler foresaw a need for a new education paradigm long ago in his bestselling book Future Shock; he stated that, "Tomorrow's illiterate will not be the man who can't read; he will be the man who has not learned how to unlearn [8]." 100 years ago, John Dewey had given some important clues regarding this paradigm shift in his seminal work Democracy and Education that "A society which is mobile, which is full of channels for the distribution of a change occurring anywhere, must see to it that its members are educated to personal initiative and adaptability. Otherwise, they will be overwhelmed by the changes in which they are caught and whose significance or connections they do not perceive [9]."

However, traditional educational approaches still dominate the world. This widespread approach in education has been named "dominant design of educational systems", with typical characteristics of using technology only within the course context, asymmetric relationships between learner and teacher, homogenous student experience, using e-learning standards, restricted access of contents and organizationally bounded scope [10]. An alternative approach called "Personal Learning Environment (PLE)" has been proposed to address the flaws inherent in the classical instructional approach, with characteristics of orchestrating connections between users and services, symmetric relationships, individualized contexts, open internet standards, open content, and intermingling of personal and global scope [10].

In this study, the following contributions are made:

- Personal learning needs have been studied using semi-structured interviews with the Information and Communication Technology (ICT) practitioners in the Turkish Software Industry Ecosystem.
- Personal learning needs have also been studied using PLE literature reviews conducted previously, and emerging technologies.
- A reference intelligent PLE Architecture has been proposed and evaluated using requirements gathered from prior art and field studies.

First, a preliminary architecture is proposed. Second, interviews are conducted. In the meantime, literature review is done. To prepare for the SAAM architecture evaluation, user scenarios are constructed using the outputs of the interviews and the literature review. SAAM evaluation is made iteratively until the resultant architecture covers all the user scenarios.

2. BACKGROUND

The term Personal Learning Environment, was first coined by Van Harmelen [11]. It is an environment in which a learner leads his/her learning process and bi-directional feedback mechanisms are maintained between the environment and the learner. This new approach represents a challenge for the existing "dominant design of education" [10]. Study in [12] authors tried to overcome this challenge by using Personal Learning Environment approach as a complementary extension on formal learning systems. Throughout the last decade, there has been a large amount of PLE studies in the literature. At least 400 papers have been published according to a recent PLE review study [13]. However review studies about PLEs are scarce. A review study working out PLEs using Activity Theory framework referenced over 100 PLE papers [14]. Another study reviewed the research trends in education and its relation with PLEs [15]. In contrast, there are quite a few review studies concerning Learning Analytics (LA), Educational Data Mining (EDM) and Academic Analytics [16]–[24]. These reviews are summarized in [25]. Accumulated knowledge in aforementioned PLE literature and field survey results will be used in generating the user scenarios outlined in detail in Section 3. These scenarios will be used in evaluating the proposed reference intelligent PLE architecture.

Software architecture evaluation is not an easy task since of the concerns of many stakeholders' are involved and optimizing the expectations poses several tradeoffs on the architecture. SAAM [26], [27] is one of the architectural evaluation methods proposed in Software Engineering literature and has been in use in the industry throughout the last two decades. It's a scenario based, simple, and efficient method. [27]. There are several other architecture evaluation methods such as ATAM (Architecture Tradeoff Analysis Method) and ALMA (Architecture-Level Modifiability Analysis) [28]. We used SAAM since it's a simple yet efficient method. It's also among the most validated methods by numerous case studies together with ATAM [28].

3. PROPOSED REFERENCE PLE ARCHITECTURE: REFPLE

In the following subsections, first, interview findings are presented. Second, a preliminary reference PLE architecture is described and validated using SAAM.

There were five requirements engineers and five software developers who participated voluntarily for the interviews. These ten participants were interviewed one on one. Interviews were held during 45 minute meetings and designed in a semi-structured manner. The interviewer asked four main open-ended questions regarding participants' views on personal learning. Some additional follow-up questions have been designed to capture more specific additional information. Main questions are as follows:

- How do you make progress on your personal learning and development?
- What challenges do you face during your personal learning activities?
- What do you think of self-regulated learning?
- What kind of emerging technologies can be utilized in personal learning in your opinion?

The recorded data during the one on one meetings was analysed using the thematic analysis method [29]. The codes and themes identified from this analysis are shown in Table 1.

Table 1 Themes and Codes after Thematic Analysis of Interview Content

Themes	Codes	Frequency
Assessment & Measurement	Measurement, Skills Gap, Development Analysis, Recognising Learner, Learner Feedback	3
Colloborative and Social Learning	Social Platforms, Forums, Stackoverflow, Community, Contribution, Reflection, Reputation, Blogs, Twitter,	25

	Teaching, Contributing, Asking Questions, Interactive, Dissemination of Success, Failure Practices, Facebook, Whatsapp, BIP (A messaging platform), Skype, Social Environment	
Constraints	Time Constraint, Environmental Restrictions, Constraints, Security Constraints	5
Ethics, Privacy and Theory	Ethics, Privacy Concerns, Pedagogy, Androgogy, Health, Technology Balance, Addiction	3
External Learning Environment	Google, Udemy, Coursera, Internet Resource, Amazon, Digital Learning, Ted, Youtube, University, Higher Education, Javacodegeeks, Learning by Reading, Kindle, W3schools, HBR, Quora, Dice.com, futurism.com, mashable.com, techcrunch.com, Hobby	51
Interoperability	RSS, Integration	2
Formal Learning	Institutional Learning, Mandatory Learning, Classroom, Instructional Learning, K12	7
Learning Resource	Video Content, Printed Resource, Knowledge Management, Light Documentation, Book, PDF, Library, English Language Efficacy, Problem, Requirement Definition, Powerpoint, Paid Content Cracking the coding interview, Book Review, Interactive Content, Tutorial	27
Learning Resource Reliability	Resource Credibility, Reliability, Quality, Trustability, Garbage Information, Author Quality, Expertize, Best Expert, Mark, Brand, Quality Resources, Content Sufficiency, Up-to-date Content	11
Learning Resource Retriveal	Search, Query Design, Quick Access to Resources, Tagging, Relevant Resource	18
Mentoring, Coaching, Scaffolding	Mentor, Coach, Expert, Teacher, Carrier Planning, Biased Teacher, Knowledge Sharing of Experts, Ego	16
Motivation	Motivation, Gamification, Certification, Incentive, Prize, Life-Work Balance, TODO List, TOBE Learned List, Simplicity	20
Learning Journey Design	Personalised Corporate Learning, Incremental Level, Staged Learning, Learning Journey Patterns, Curriculum	5
Practice & Simulation	Learning by Practice, HackerRank, Learning By Problem, Case, Project, Pluralsite, Practice, Simulation, Virtual, IAAS Environments, Experimental, Discovery Learning	12
Technologies in Learning	Learning Assistant, Virtual Reality, Face2Face, Videoconference Learning, Mobility, Augmented Reality, AI, Personal and Institutional, Instructional Learning Balance, Big Data, Just in time learning, Vertical vs Horizontal Learning, Specialisation, Human vs AI, Ad hoc learning, Adaptive, Contextual	37
Recruitment Platforms	LinkedIn, Living	3
Workplace Learning	Workplace Learning, On the job, Embedded Learning, Workers Learning, Bag Farming, Formal Learning, Informal Learning, Higher Education, Support, benefits in parallel to Workplace Learning, Traversal/Network Learning	13

Modules and structure of the first version of the architecture are based on the architecture proposed in [25]. The functional viewpoint of the architecture is shown in a "boxes-and-lines

diagram" notation [30]. Explanations for the functional components comprising the architecture can be found in [25].

User scenarios are generated with all the accumulated knowledge in previous sections. The scenarios have been used in evaluating the reference PLE architecture using SAAM. Main references in generating these scenarios are [13]–[24], [31]–[36]. These use cases covers a wide spectrum of requirements from "predicting learner's failure and success" to "addressing prvacy, trust and ethical concerns".

Modules and structure of the final version of the architecture which has been enhanced by the evaluation applied in previous section are shown in Figure 1. There are several new and revised architectural components in the enhanced architecture compared to the first version of the architecture. The core components that enable the architecture to be intelligent are Real Time Learning Analysis, Assessment and Measurement, Learning Resource Retrieval, Learning Resource Reliability, PLE Design, PLE DB, Lambda Big Data DB, Data Smoothing and Interface Adaptor modules. These components are shown in dashed boxes in Figure 1.

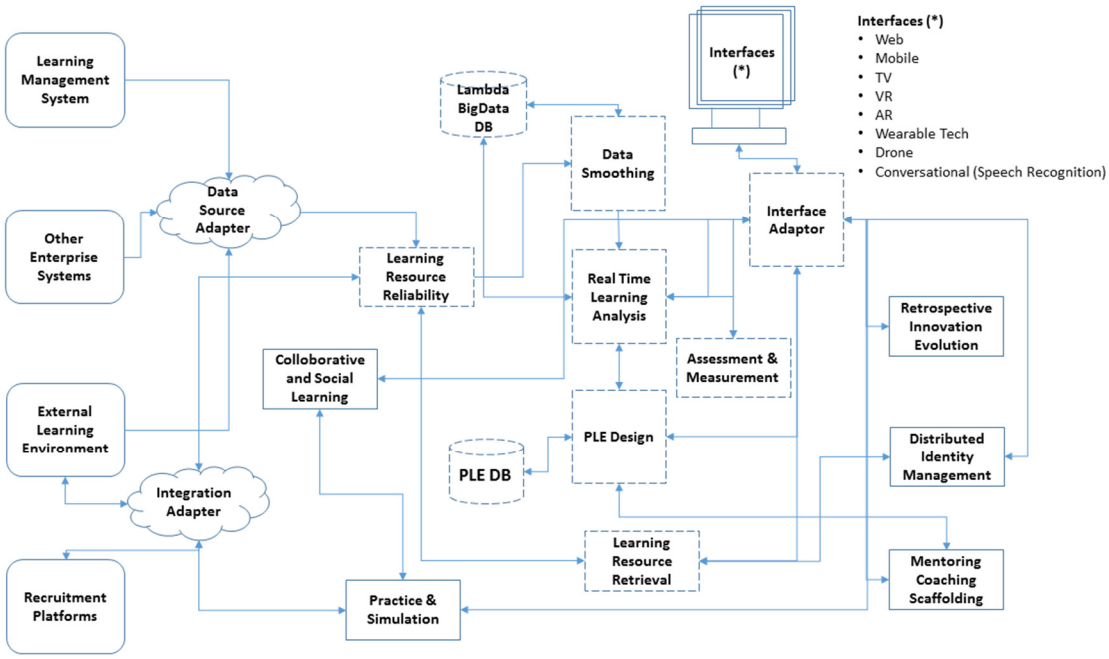

Figure 1 REFPLE Architecture After Evaluation

4. DISCUSSION

We designed and validated intelligent personal learning environment architecture for the 21st century learning. The architecture has necessary components to assist lifelong learners in their up skilling and reskilling process to cope with the challenges of the current era.

Personal learning environment is a relatively new approach. Hence there are a small number of proposals for its technical architecture in literature. The proposals by [10], [37]–[41] are very high level architecture, frameworks and approaches. Although they cover the general PLE concept, they lack technical component details and rich functional scope. [42] Proposes a more refined conceptual architecture called iPLE which merges personal and institutional learning. In relatively new research [43], a design framework is proposed and also pedagogical aspects are studied.

Another major limitation of existing proposals is that they don't use machine learning, big data analysis and other emerging technologies. Therefore, to a large extent, previous proposals

are not intelligent systems. There are just a few exceptions. [44] Designs a recommender system within a PLE architecture using semantic web technologies, ontologies and collaborative tagging. [45] Proposes a personalized learning approach for learners with multiple disabilities using some basic machine learning technics. Apart from this approaches, to our knowledge, the only proposal PLE architecture that makes use of machine learning is [46] in literature. Additionally, although they are not a PLE architecture proposal per se, [47] introduce an intelligent recommender mechanism within a Learning Management System using ontology and semantic web; [48] proposes an intelligent web learning environment than can be personalized to learners; [49] used some machine learning algorithms to identify learning patterns. This study differs and makes contributions in the following ways:

Functional coverage of the architecture is very broad since it takes into account the requirements in the PLE literature accumulated in the course of the last two decades and up-to-date needs of practitioners in workplace captured during field studies conducted in this research. It has big data analysis and artificial intelligence capabilities as well as some other emerging technologies such as virtual reality, augmented reality and conversational interfaces. It is the first PLE architecture which is validated using a Software architecture evaluation method.

The architecture has been evaluated using field studies conducted within the ecosystem of a large telecommunications company in Turkey and the İstanbul district in particular. Another constraint is the target professionals chosen during the field studies. They don't represent all professionals but only a subset of them, which is ICT practitioners. Although architecture evaluation has been done using detailed requirements from personal learning environment literature as well, this may not be sufficient for generalizability of the findings.

5. CONCLUSION AND FUTURE RESEARCH

In this study, we conducted a multi-stage research to design and validate a reference architecture for intelligent personal learning environments. First, we carried out semi-structured interviews with employees of a large ICT company concerning their personal learning views and needs. Interview material has been analyzed using thematic analysis to construct codes and themes. Second, using the output from thematic analysis, an online survey was designed and 166 employees from the ICT Company and its ecosystem companies participated in the survey. Finally, a reference architecture for PLE was designed, validated and extended through SAAM. Use cases employed in the SAAM evaluation were derived from the information captured from interviews, survey, and literature reviews.

We believe that the proposed architecture will be a useful reference material for researchers, companies that develop products on learning technologies, institutions that have the mission of addressing the future skills gap, and all life-long learners. In the future, intelligent PLE architecture's use cases containing different dimensions of the workplace environment will be studied further.

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